Forecasting HDB Monthly Rents with Historical Cohorts and Heterogeneous Ensembles

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***Abstract*—Estimating residential rents requires separating property differences from a changing market while respecting the information available at prediction time. We study monthly rental prediction for Singapore Housing and Development Board flats using 150,000 labelled transactions and supplied building, amenity and economic data. Our experiments separate model comparisons with common inputs from controlled feature ablations and comparisons of complete pipelines. On the annual development window, XGBoost achieves a root mean squared error of 467.03 Singapore dollars with 27 common inputs. Adding ordinary, market-relative and anniversary rental histories reduces a fixed LightGBM model's error from 481.60 to 457.84. A grouped tree/Transformer pipeline records annual and recent development errors of 451.03 and 441.83. In a separate recent-window follow-up, anniversary histories reduce frozen TabPFN error from 455.40 to 441.96; an equal blend with a seven-tree ensemble reaches 440.95, or 440.77 with coarser flat-model categories. Filtering unusually low rents worsens squared error. Repeated reuse of development periods and limited future-horizon coverage constrain generalization claims. The main finding is that carefully bounded historical information contributes more consistently than additional model complexity.**

***Index Terms*—HDB rental prediction, temporal validation, feature engineering, gradient boosting, tabular Transformers, ensembles**

I. Introduction

An estimate of a flat's monthly rent can help prospective tenants compare listings, help landlords set realistic expectations, and support analysis of housing affordability. A useful estimate must reflect both the dwelling and the rental market at the intended date. Predicting a typical rent for a location is insufficient when prices change substantially over time or when superficially similar flats rent at different levels.

We predict the monthly rent, in Singapore dollars (SGD), of an individual Housing and Development Board (HDB) rental record from its approval month, property attributes, address and supplied auxiliary information. This is a rental task, despite the repository's broader "price prediction" name. Our objective is reliable prediction and interpretation of its limitations, assessed primarily through chronological validation.

Three challenges drive the methodology. First, rents shift over time, so a random train–validation split could reward interpolation between contemporaneous transactions rather than forecasting. Second, area, flat type and location overlap in the information they encode; feature importance alone cannot identify their independent predictive contribution. Third, the data omit unit-specific factors such as actual floor, interior condition and contract details. Two records with the same available attributes can legitimately have different rents.

We ask: (1) how much do static spatial features and earlier rental histories add under a controlled model; (2) do neural networks outperform tree methods when given the same information; and (3) can heterogeneous predictions improve accuracy without concealing weak transfer or information leakage? The study's contributions are a time-bounded history representation, controlled model and feature comparisons, and an assessment of conditional blending and negative results. We distinguish the archived grouped pipeline from subsequent TabPFN experiments and compare their results within the appropriate validation window.

A. Related work and scope

Sing, Yang and Yu [1] apply boosted-tree automated valuation models to Singapore's public and private housing transactions, providing a local precedent for nonlinear property valuation. Their target is sale value; we study monthly rent under a changing market and do not transfer their reported errors to this task. Broader tabular benchmarks by Grinsztajn, Oyallon and Varoquaux [2] motivate strong tree baselines alongside neural models, but do not determine performance on our larger rental dataset. Our contribution is empirical: compare model families under common inputs, isolate the increment from historical cohorts, and examine whether ensemble gains transfer between chronological origins. The histories are predictive summaries, not causal estimates of property attributes or verified renewal links.

II. Data Understanding and Preprocessing

A. Dataset and exploratory findings

The supplied training table contains 150,000 records from January 2021 to March 2025, with ten predictors and the target `MONTHLY_RENT`. The 50,000 unlabelled test records span March 2025–July 2026. Predictors include approval month, town, block, street, flat type, flat model, floor area, furnished status, lease commencement year and fee. The test period therefore overlaps the final training month and extends well beyond it.

Raw CSV cells have no missing entries, but semantic cleaning is necessary. There are 425 exact duplicate rows; the principal experiments retain them because the available fields do not identify individual contracts and repetition alone does not establish an error. `FURNISHED` and `FEE` are constant and

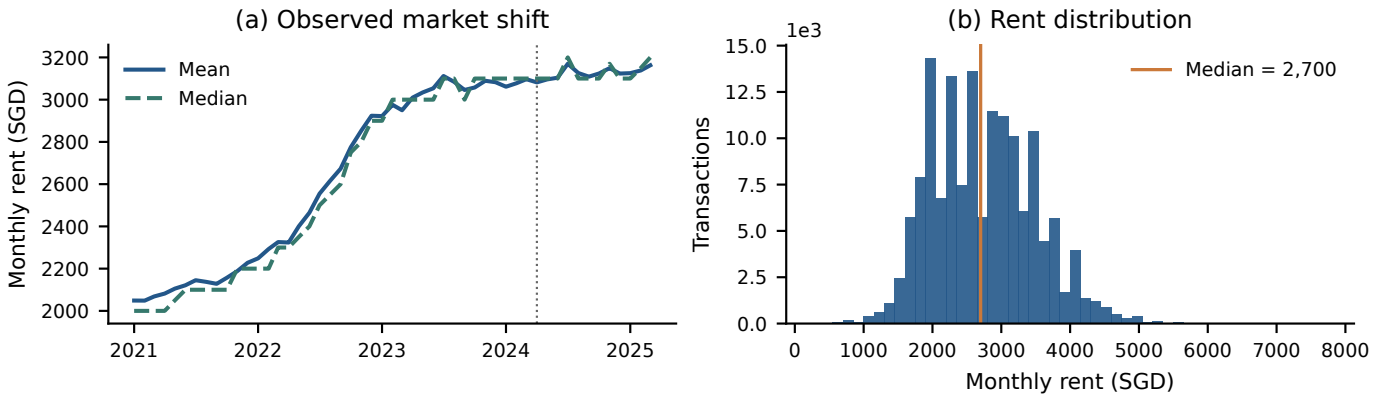


Fig. 1. Training-data EDA. Monthly means and medians show substantial temporal change; the dotted line marks the annual validation start. The marginal distribution spans S$300–S$7,600. These are descriptive summaries of the supplied training data, not composition-adjusted price indices.

are excluded. Flat types contain equivalent spellings such as "4 room" and "4-room". The naming convention changes strongly with year, so standardizing case, whitespace and separators also removes a formatting proxy for time.

After normalization, the largest groups are 4-room (54,079), 3-room (49,024) and 5-room flats (35,523); only 42 observations are 1-room flats. The median rent is S$2,700, the interquartile range is S$2,100–S$3,200, and the 1st–99th percentile range is S$1,300–S$4,600. Figure 1 motivates explicit calendar and market-history features. Area has a raw Pearson correlation of approximately 0.329 with rent, suggesting that size alone cannot capture temporal and spatial variation. These associations are descriptive and do not establish causal premiums.

B. Auxiliary information and cleaning decisions

Six supplied auxiliary tables describe HDB blocks, MRT stations, malls, schools, Certificate of Entitlement (COE) premiums and stock prices. The building table has 9,464 entries. Normalized town–block–street keys match all train and test records in the existing auxiliary audit. Postal code "NIL" and invalid numeric sentinels such as -1 are treated as missing rather than valid values. One training record has a lease commencement year after approval; its negative age proxy is set to missing.

Station records must be deduplicated by physical location: 243 line-level entries represent 192 distinct locations in the supplied audit. Otherwise, interchange stations inflate nearby-station counts. Coordinates support nearest-facility great-circle distance and counts within 1 km. These describe geographical proximity, not measured travel time. Building maximum floor describes building form and must not be interpreted as the rental unit's floor.

The facility tables are static snapshots without complete historical opening dates. We therefore treat them as competition-supplied spatial information, not as fully audited historical inventories. This limits claims about deployable point-in-time accuracy. Similarly, economic series observed after a forecast origin are unavailable in a realistic fixed-origin forecast. Experiments using supplied prediction-month economic observations are explicitly reported as a different information scenario.

Missing numerical history values remain missing for models that support them; counts are zero when a group has no observations. Linear and neural pipelines learn imputation and scaling only on fitting rows. Categorical vocabularies are also learned within each fitting partition, with a reserved unknown category. Neural preprocessing includes categorical embeddings, log-transformed history counts and, where configured, train-fitted numerical transformations. Continuous time retains an extrapolatable linear scale rather than being quantile-clipped at the last observed month.

C. Outliers and duplicate handling

We do not remove rents merely because they are small or large. A low rent may reflect a valid contract, omitted unit information or a data problem; those explanations cannot be distinguished from the target alone. The main results preserve every validation record. A later training-only half-median filter tests whether removing unusually low rents helps; Section V-E reports its deterioration. This makes cleaning a testable modelling decision rather than a way to simplify the evaluation distribution.

III. Feature Engineering

A. Static and ordinary historical features

The compact static representation has 19 inputs: five property variables, three calendar variables, five location variables and six amenity variables. Property inputs include normalized type and model, floor area, age proxy and building maximum floor. Calendar inputs are month index and annual sine/cosine terms. Location inputs include town, planning area, latitude, longitude and distance to the chosen CBD reference point. Each amenity family contributes nearest distance and count within 1 km.

Eight history fields extend this to 27 inputs. At both block $\times$ flat-type and planning-area $\times$ flat-type levels, we calculate expanding mean rent, observation count, latest observed

monthly mean and months since the latest observation. For a fitting record in month t, these statistics use only months earlier than t. For a validation record, the source pool is frozen before the forecast origin T_0:

$$\mathcal{H}_i = \{j : g_j = g_i,\ t_j < \min(t_i, T_0)\}, \tag{1}$$

where g denotes the relevant group. No validation rent updates the pool. For fitting records, $t_i < T_0$, so Equation 1 reduces to a strictly earlier-month lookup.

This representation supplies both a local price estimate and its reliability. Counts distinguish sparse estimates from well-supported averages; recency distinguishes stale and current information. Unknown block groups can fall back indirectly to area-level information through the model. Four signature-history columns were removed from an earlier 31-column representation after a rowwise audit showed that they duplicated block/type histories in this dataset.

B. Market-relative and anniversary histories

To separate a location's price level from the national market at the source date, we normalize a historical rent by its source-month flat-type mean:

$$q_j = \frac{y_j}{\overline{y}_{f_j, t_j}}. \tag{2}$$

Twenty summaries over five grouping granularities extend the 27-column representation to 47 columns. Because normalization uses historical source-month rents, it does not require a target-month market mean at inference.

Anniversary features then retrieve same-block, same-flat-type observations around $t_i - L$, for $L \in \{12, 24, 36\}$ months, with a ± 1-month window. Each lag contributes five quantities: sample count, mean rent, mean market ratio, standard deviation of the ratio and fraction with ratio below 0.65. The resulting nested representation has 62 columns. Exact-month and broader-cohort variants are also used by ensemble members. All retrieved records must lie in the fitting pool; unavailable cohorts remain missing with zero support.

The hypothesis is that cohort-specific historical levels preserve information diluted by an expanding average. They might partly reflect recurring rental patterns, but no unit or tenant identifier is available. We therefore do not claim to identify renewals. At long horizons, an anniversary month can lie beyond the frozen source pool; a correct implementation leaves that cohort unavailable.

C. Local temporal baselines and residual targets

Some enhanced models learn residual rent relative to a baseline b_i built exclusively from earlier information:

$$r_i = y_i - b_i, \qquad \widehat{y}_i = b_i + \widehat{r}_i. \tag{3}$$

The baseline is added exactly once before comparing models or blending predictions in SGD. The submitted Transformer branch uses an adjusted historical baseline. The separate aligned TabPFN branch pools flat-type trends and shrunken planning-area premiums, selecting damping and decay using earlier rolling forecasts.

For TabPFN, history features and the temporal baseline share the same simulated origin for each context task. This prevents a training mismatch in which stale forecasts are paired with fresher local histories. Context tasks cover the one-to-ten-month horizons used in recent validation. The TabPFN 77-input representation is a separate feature system; its count must not be interpreted as another step in the 19/27/47/62 ladder.

IV. Methods and Experimental Protocol

A. Chronological evaluation and comparison scope

The main annual window is April 2024–March 2025: 114,842 fitting rows precede 35,158 validation rows. The recent window is June 2024–March 2025: 121,005 fitting rows precede 28,995 validation rows. These windows overlap and cannot be counted as independent replications. A 16-month December 2023–March 2025 stress window, and earlier chronological windows, are used in selected studies; stress-window errors are kept separate from annual pooled results.

For the common-input model comparison, October 2023–March 2024 is the inner selection period. Candidate hyperparameters and stopping rounds are chosen there; the selected configuration is refitted on all pre-April rows for a fixed number of rounds. Outer labels do not drive early stopping. This follows the forecasting principle that evaluation targets must occur after the information used to construct their forecasts [3]. Later blend experiments have their own chronological meta-training cutoffs, described below.

We distinguish three questions. The *common-input comparison* fixes the 27 source features and fitting/validation rows. The *feature ladder* fixes each model's parameters and rounds while adding nested feature groups. The *pipeline comparison* allows enhanced representations and blending. Differences between pipelines cannot be attributed solely to architecture.

The primary metric is original-scale root mean squared error:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\widehat{y}_i - y_i)^2}. \tag{4}$$

We also report mean absolute error (MAE) and mean signed bias, $n^{-1}\sum_i(\widehat{y}_i - y_i)$. Positive bias means overprediction. Where disjoint folds are pooled, their summed squared errors and row counts are pooled before taking the square root; fold RMSEs are not averaged directly.

B. Model families and tuning

Ridge regression provides a linear reference, with $\alpha \in \{1, 100, 10000\}$ selected on the inner window. ExtraTrees tests randomized bagging. LightGBM [4], XGBoost [5] and CatBoost [6] test strong nonlinear tabular baselines with different handling of sparse and categorical inputs. Embedding MLP and FT-Transformer [7] test whether learned category representations and feature attention improve on trees.

The archived search uses a limited candidate set rather than exhaustive tuning: LightGBM tests 31/63 leaves, CatBoost depth 6/8, and XGBoost depth 4/6, alongside representation

variants. Common-input comparisons restrict eligibility to the compact profile. Inner-selected training lengths are 108 LightGBM trees, 448 XGBoost rounds, 814 CatBoost rounds, 300 ExtraTrees trees, 29 MLP epochs and five FT-Transformer epochs. The MLP uses width 256 and learning rate 10^{-3}; the compact FT uses width 96 and learning rate 3×10^{-4}. Search budgets and preprocessing differ across families, so the comparison is controlled for information rather than total compute.

The feature ladder has a separate, fixed configuration: LightGBM uses 353 trees, 15 leaves, learning rate 0.03 and L2 penalty 10; CatBoost uses 535 iterations, depth 6, learning rate 0.04 and L2 penalty 10. These settings stay constant across all four feature levels within each model. Their 27-input errors therefore need not equal those of the separately tuned common-input models.

C. Numerical embeddings and structural alternatives

The enhanced Transformer study compares linear numerical tokens, piecewise-linear embeddings (PLE), and learnable periodic embeddings (PLR) [8]. The selected PLR-FT member has width 128, three attention blocks, eight heads and learning rate 10^{-4}. It uses 16 periodic frequencies per numerical feature. Training uses AdamW with weight decay 0.001; the earlier-window epoch cap is 16 with patience four. Later refits use the selected epoch count without validation stopping.

New neural members omit five sparse 36-month cohort statistics and learn residuals relative to the adjusted history baseline. A four-seed average of the pruned FT is one ensemble channel. Additional bounded-budget alternatives include a DCN-V2-style full-matrix cross network [9], soft-gated expert networks and asymmetric CatBoost trees. These are project adaptations, not claims of reproducing published benchmark results. Their negative and mixed results inform the final selection.

D. Regularized blending conditioned on history coverage

The archived final pipeline contains 12 prediction channels: seven historical tree members, two XGBoost members, the original FT, a four-seed pruned-FT average and a PLR-FT member. Thus it contains 15 trained predictors, not 15 architectures. All outputs are first reconstructed as SGD rents.

Let $e_{ik} = (p_{ik} - y_i)/1000$ be model k's historical out-of-time error. For blend fitting, each model's within-month weighted mean error is subtracted to obtain $\widetilde{e}_{ik}$. Global weights solve

$$\begin{aligned} \min_{w} \quad & \sum_{i} a_i (\widetilde{e}_i^{\mathsf{T}} w)^2 + \lambda \|w - w^{(0)}\|_2^2, \\ \text{subject to} \quad & \mathbf{1}^{\mathsf{T}} w = 1, \quad 0 \le w_k \le 0.6, \end{aligned} \tag{5}$$

with $\lambda = 0.003$ and prior $w^{(0)}$ from the preceding ensemble. Dividing errors by 1,000 is part of the penalty's scale. Month-centering aims to reduce sensitivity to historical market-level bias. It is used only during weight estimation; inference directly averages predictions and never needs a query-month error mean.

TABLE I
COMMON 27 INPUTS; 35,158 ANNUAL VALIDATION ROWS. ALL ERRORS ARE IN SGD. BIAS IS PREDICTION MINUS ACTUAL RENT.

Model	RMSE	MAE	Bias
Ridge	513.03	363.23	+194.11
ExtraTrees	468.47	341.08	+4.79
LightGBM	468.87	340.71	+11.67
XGBoost	467.03	338.87	+20.64
CatBoost	468.46	343.51	-3.71
Embedding MLP	483.72	358.21	-11.83
FT-Transformer	476.00	339.92	+87.64

The presence or absence of 24- and 36-month cohort support defines at most four groups. Group weights solve the same problem with the global solution as prior and penalty 0.003. Groups with fewer than 500 meta-training observations, and unseen groups, use global weights. Annual blend labels stop at March 2024; recent weights may include April–May out-of-time predictions made by pre-April models. The final full-data blend uses 53,135 historical out-of-time predictions from October 2023–March 2024 and April 2024–March 2025, weighting the two forecast-origin pools equally.

The month-centered grouping recipe and the decision to include PLR were selected after examining later development results. Chronological weight fitting prevents direct use of a query's label in its own forecast, but does not erase this adaptive model-selection bias.

V. EVALUATION AND INTERPRETATION

A. Common-input models

Table I shows that the tested tree models outperform the tested MLP and FT configurations on annual RMSE. XGBoost has the smallest error, S$467.03, compared with S$513.03 for Ridge and S$476.00 for FT. The training-mean benchmark is much worse at S$846.75. Nonlinear models therefore offer a substantial improvement over these simple references.

However, differences among the tree methods are small: the XGBoost–LightGBM gap is only S$1.84. The archived three-month-block sensitivity interval for this gap is approximately $[-2.48, -1.28]$, conditional on the saved predictions. This is not an architecture-wide superiority claim. Single seeds, unequal search budgets and reuse of the annual period limit that interpretation.

FT's MAE of S$339.92 is close to the strongest trees despite a larger RMSE and positive bias. This suggests that its error distribution differs from theirs and motivates checking complementarity, but a different bias or lower error correlation alone does not guarantee that adding it improves a blend.

B. Controlled feature contributions

The fixed-model ladder in Table II and Figure 2 supports two clear gains. Adding ordinary histories reduces annual LightGBM RMSE by S$13.81, from 481.60 to 467.80. Adding anniversary cohorts to the 47-input model reduces it by another S$9.12, to 457.84. CatBoost shows the same direction,

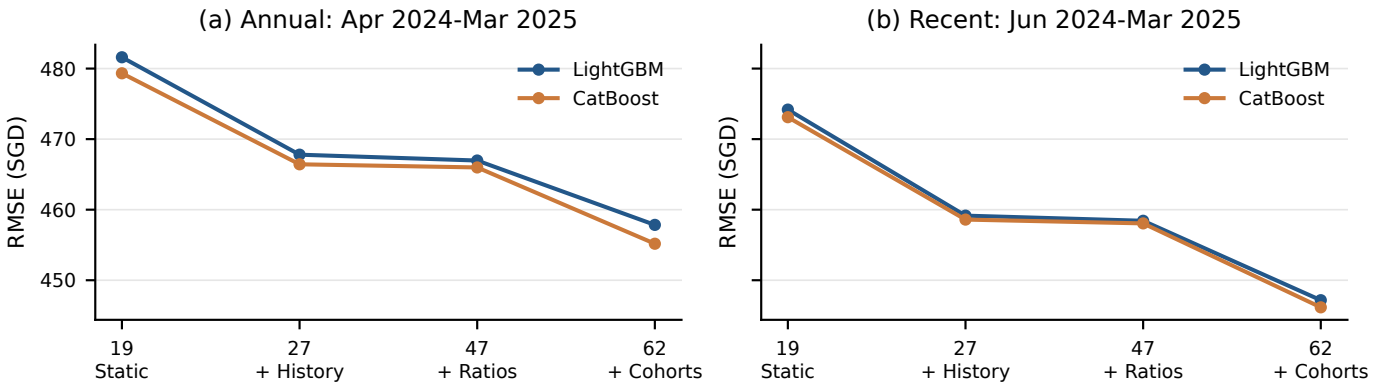


Fig. 2. Controlled nested feature ladder. Parameters, training rounds and evaluation rows are fixed within each model/window. Ordinary histories and anniversary cohorts provide clearer gains than the intermediate market-ratio expansion. Annual and recent windows overlap.

TABLE II
FEATURE LADDER RMSE (SGD). COUNTS DENOTE NESTED INPUT SETS, NOT CHANGES OF MODEL ARCHITECTURE.

Model / window	19	27	47	62
LightGBM / annual	481.60	467.80	466.97	457.84
LightGBM / recent	474.18	459.15	458.43	447.17
CatBoost / annual	479.33	466.43	465.99	455.17
CatBoost / recent	473.11	458.60	458.06	446.16

ending at 455.17. Market-relative histories alone yield smaller increments: approximately S$0.83 for annual LightGBM and S$0.45 for annual CatBoost.

Group-deletion experiments provide a complementary check. On the compact 27-input LightGBM model, removing all history fields raises pooled three-annual-window RMSE by S$7.94, while removing time and all histories together raises it by S$141.33. Removing all amenity fields raises pooled error by only S$0.58, with inconsistent window-level effects. Predictors can substitute for one another, so single-column deletions are not additive importance shares. Retained histories may still encode location or flat type when direct location/type inputs are removed.

The small conditional amenity gain does not establish that amenities are economically unimportant. Location and past rents may already absorb much of their predictive information, and static facility snapshots are imperfect proxies. The evidence supports prioritizing time and history engineering under this evaluation design.

Supplementary Figure 4 visualizes the deletion results and the availability of anniversary histories. In the TabPFN cohort study, 36-month support is available for only 8.5% of fitting records, compared with 63.2% of validation records. This support mismatch is a reason to inspect sparse long-lag features carefully, rather than assuming more history always improves transfer.

C. Enhanced pipeline validation and structural ablations

Table III compares selected complete pipelines, not the best standalone result of every model family. Their feature sets and blending rules differ from the common-input configurations in Table I. The tree/Transformer pipeline improves on the history ensemble on both windows; recorded fixed-recipe refits reproduce its annual/recent RMSE of 451.0295/441.8345. The later tree/TabPFN candidates have lower recent-window errors, but were not evaluated on the annual window. Their recent advantage therefore does not establish superiority across forecast origins.

The supplied-macro variant changes recent RMSE by only S$0.0133 and slightly worsens annual RMSE. Because it uses prediction-month economic observations, its separate information scenario and archived results are retained in Appendix A.

The final-versus-history gain combines several changes: history pruning, seed averaging, conditional blending and PLR output. It cannot all be credited to PLR. Holding the grouped-blending rule fixed, adding only PLR changes recent RMSE from 441.9591 to 441.8345, but worsens annual RMSE from 450.8650 to 451.0295. The improvement is therefore not uniform across origins.

PLR-FT is also not the best standalone network: PLE-FT records annual/recent errors of 458.29/449.82, versus 462.07/451.10 for PLR-FT. PLR's value is conditional complementarity within this blend. Adding every new architecture instead yields recent RMSE 442.51, worse than the grouped baseline. Increased model count does not guarantee useful diversity, and bounded weight optimization can still overfit historical error patterns.

D. Separate TabPFN follow-up

The foundation-model branch uses TabPFN-3.5-Fast [10], full 121,005-row context, seed 42 and two internal estimator views. In a matched ablation, its original 77 inputs, residual target, temporal baseline and checkpoint stay fixed. Adding ten 12/24-month cohort inputs reduces RMSE from 455.396 to 441.959 (Table IV), a S$13.437 gain. Five additional 36-month fields yield 442.016, providing no further observed RMSE improvement.

Combining the 87-input TabPFN with the reproduced seven-tree ensemble at equal weights yields 440.952. A TabPFN

TABLE III
SELECTED PIPELINE RMSE (SGD). ANNUAL: 114,842 FIT THROUGH MARCH 2024 / 35,158 VALIDATION IN APRIL 2024–MARCH 2025. RECENT: 121,005 FIT THROUGH MAY 2024 / 28,995 VALIDATION IN JUNE 2024–MARCH 2025. OVERLAPPING DEVELOPMENT WINDOWS; – MEANS NOT EVALUATED.

Pipeline	Annual RMSE	Recent RMSE
History ensemble	451.1810	442.9025
Tree/Transformer + PLR-FT	451.0295	441.8345
Trees + TabPFN, original categories	–	440.9520
Trees + TabPFN, grouped flat models	–	**440.7707**

TABLE IV
SEPARATE RECENT-WINDOW FOLLOW-UP: FROZEN TABPFN-3.5-FAST AND THE REPRODUCED SEVEN-TREE ENSEMBLE. THIS TREE ENSEMBLE IS AN EARLIER MODEL, NOT THE SUBMITTED 12-CHANNEL FINAL.

Model	Inputs	RMSE	MAE
Frozen Fast control	77	455.396	327.794
+ 12/24-month cohorts	87	441.959	318.216
+ 12/24/36-month cohorts	92	442.016	317.815
Seven-tree ensemble	varied	443.863	322.919
Earlier-locked blend	varied	442.302	321.032
Equal blend	varied	440.952	318.888

TABLE V
EXACT GROUPED-FLAT-MODEL ENSEMBLE MEMBERS ON THE RECENT SPLIT. ERRORS ARE IN SGD; ROUNDED WEIGHTS ARE PERCENTAGES OF THE FINAL BLEND. LGB SUFFIXES DENOTE LEAF COUNTS; CAT6 DENOTES DEPTH SIX. THE LAST TWO ROWS ARE SUMMARIES.

Member	Weight (%)	RMSE	MAE
Market-ratio LGB15	6.547	471.727	350.746
Broad LGB15	4.829	448.308	325.268
Broad LGB63	5.176	451.190	327.327
Exact LGB15	8.655	449.358	326.930
Exact LGB63	0.329	448.908	325.500
Window Cat6	20.690	446.603	325.256
Window LGB63	3.773	447.235	323.823
Frozen TabPFN87	50.000	442.089	316.516
Seven-tree subtotal	–	443.956	322.957
Combined ensemble	–	**440.771**	317.762

weight of 0.1893 fitted on October 2023–March 2024 forecasts instead yields 442.302 on the recent window. Thus the attractive equal-blend result should be read alongside the less favorable earlier-locked choice. It improves all ten months relative to TabPFN and nine relative to the tree ensemble, but the same months have already informed multiple experiments.

Replacing direct flat-model categories with a proposed coarser grouping yields a further small blend decrease to 440.771, but its descriptive month-cluster interval includes no improvement and errors worsen among the 126 validation rows whose labels change. This is insufficient to establish a robust grouping benefit. We therefore emphasize the matched cohort ablation and the ungrouped equal-blend result, while retaining the coarser-group result as a sensitivity analysis requiring confirmation at other chronological origins.

Table V reports the exact enhanced members, distinct from Table I's common-input models. Six LightGBM models and one CatBoost model share half the final weight; frozen TabPFN receives the other half. All use the same recent split, with tree rounds and blend weights fixed before refitting. Broad cohorts include block, street and area histories; exact/window variants use block cohorts with zero/one-month radii. The locked weights are not standalone-performance rankings or feature importances.

Category preprocessing merges Type S1/S2 into Type S and three maisonette subtypes into maisonette, reducing 20 labels to 16. Train/query mappings match; vocabularies use fitting rows only. Historical keys and all rows are retained. Both grouped component blocks have slightly worse RMSE than their original versions, yet the blend improves by S$0.1813: conditional complementarity, not stronger standalone accuracy.

TABLE VI
TRAINING-ONLY OUTLIER-FILTER ABLATION ON THE SEPARATE EQUAL TREE/TABPFN BLEND. VALIDATION IS UNCHANGED. ERRORS ARE IN SGD.

Model	RMSE	MAE	Bias
All fitting rows	440.95	318.89	-0.40
Half-median filter	445.84	316.56	32.77

E. Error patterns and training-only cleaning

Figure 3 shows that error varies by month and by rent level. For this same equal tree/TabPFN blend, 1,643 validation records (5.7%) have actual rent at most S$2,000, yet account for 32.8% of squared error. Their RMSE is S$1,061.54 with positive bias S$935.91. Among 1,998 records above S$4,000, RMSE is S$607.08 and bias is negative S$466.52. Supplementary Figure 5 compares these groups' sample and error shares. These are post-hoc target groups; true rent must not be used to choose a correction at inference.

The later half-median ablation removes fitting rents below half their year × town × normalized-flat-type median, computed only within the fitting partition. It removes 848 of 121,005 fitting rows and rebuilds all affected histories, market normalizers and temporal baselines. Validation remains all 28,995 original rows, and model hyperparameters and blend weights stay fixed.

Filtering increases RMSE by S$4.89 and improves only one of ten months (Table VI). MAE falls slightly, but positive bias rises to S$32.77. Of the supported validation groups, 189

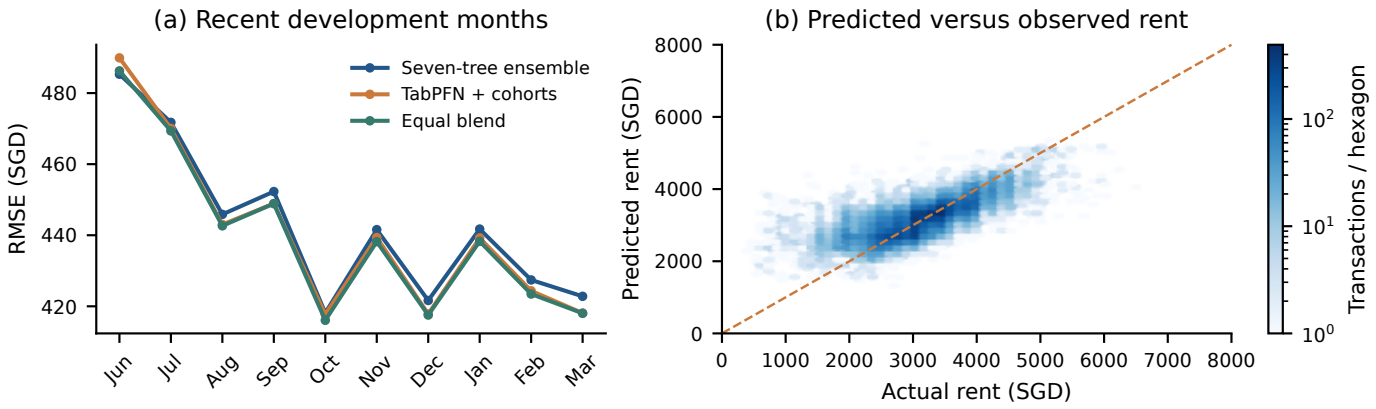


Fig. 3. Diagnostics for the separate equal tree/TabPFN blend, not the archived submitted ensemble. Left: performance varies by month and market regime. Right: log-density hexagons show regression toward central rents, with substantial overprediction of some low-rent records. The dashed line is perfect prediction.

records lie below the training-derived cutoff; their unfiltered RMSE is S$1,751.04 and rises to S$1,860.71 after filtering. The filter improves neither the primary objective nor the difficult tail. The 2025 rows lack a same-year fitting cutoff, illustrating another limitation of this retrospective rule.

These results show why removing statistical outliers is not equivalent to repairing verified label errors. The target contains meaningful heterogeneity that the observed inputs may not explain. More useful next steps are auditing source reliability, adding permissible unit-level information, and modelling uncertainty rather than automatically discarding low rents.

VI. Limitations and Reproducibility

A. What the evaluation supports

All development periods were examined repeatedly. Earlier-only features, train-fitted preprocessing and pre-origin weight cutoffs prevent direct leakage, but do not remove adaptive selection bias. Later errors informed blend selection; displayed Kaggle scores also informed the archived final choice. Small gains are development evidence, not confirmatory significance.

Paired month/block resampling retains shared market shocks, unlike an independent-row bootstrap. However, ten or twelve months provide limited temporal support, and fixed-prediction intervals omit fitting, selection and future-regime uncertainty.

Appendix B uses paired three-month blocks and recomputes pooled RMSE per resample. At one origin, monthly errors confound horizon with calendar conditions; they cannot isolate horizon effects.

The test spans 17 calendar months and includes horizons longer than the ten-month recent validation. In the full-data TabPFN policy, 18,077 test records lie beyond that validated range; the inherited temporal level and uncertainty plateau at month ten. This is a pragmatic extrapolation policy, not evidence of accuracy at those horizons. The archived 16-month stress tests cover only selected earlier pipelines and cannot validate every final configuration.

Full refits use all 150,000 supplied labels, including March 2025, while test records also start in March. This labelled-context setup differs from a strict pre-March deployment forecast: feature cutoffs and fitted-model labels are distinct boundaries. A deployment evaluation must exclude every label unavailable at its fixed origin.

B. Reproduction and remaining evidence gaps

The repository separates preprocessing, features, evaluation and fitting. Workflows record source hashes, row IDs, schemas, iterations, predictions and weights; regression checks cover temporal boundaries, reconstruction, alignment and blend constraints. Appendix A gives ownership placeholders and reproduction commands.

The report builder verifies split sizes, duplicate counts, prediction hashes, IDs, dates and labels, then recomputes recent errors. Fixed-recipe tree/Transformer refits reproduce annual/recent RMSE. Both tree/TabPFN category variants have fresh full-data refits and 50,000 aligned test predictions; all 28 validation/full-data tree fits passed checkpoint-reload checks. Document rebuilding does not retrain. New CSVs have no observed Kaggle scores; archived screenshots are not independently verified here.

VII. Conclusion

Time-bounded histories provide the most consistent gains: ordinary and anniversary features improve both LightGBM and CatBoost. Conditional blending and numerical embeddings add smaller, mixed gains. The tree/Transformer pipeline reaches annual/recent RMSE 451.03/441.83; the tree/TabPFN blend reaches recent RMSE 440.95, or 440.77 with grouped flat models. These reused development results require confirmation at untouched origins, especially longer horizons. Future work should prioritize that evaluation, source-quality audits and permissible unit-level attributes.

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APPENDIX A
TEAM WORKLOAD AND REPRODUCTION NOTES

Team details are placeholders. Insert the confirmed team name, every member's name and student ID in the author block, and replace the ownership entries below with the actual division of work. The table lists project activities; it does not assert who performed them or invent contribution percentages.

TABLE VII
WORKLOAD BREAKDOWN TO BE COMPLETED BY THE TEAM. MULTIPLE MEMBERS MAY SHARE AN ACTIVITY.

Activity	Member(s), student ID(s)
EDA, source checks and preprocessing	To be supplied
Spatial and rental-history features	To be supplied
Tree models and chronological evaluation	To be supplied
Transformer / TabPFN experiments	To be supplied
Ensembling and submission verification	To be supplied
Report, visualizations and reproducibility	To be supplied

A. Supplementary competition record

Validation results are the primary evidence in this report. For completeness, the archived November 5, 2026 selection record lists Kaggle displayed scores of 415.86912 for the history ensemble, 415.78232 for the macro variant, and 415.37529 for the grouped ensemble with PLR-FT. Its source screenshot is unavailable here and does not identify the public/private partition. These figures are supplementary archival context, not independently verified test results or a replacement for the chronological comparisons.

TABLE VIII
ARCHIVED PIPELINE COMPARISON MOVED FROM THE MAIN TABLE. ANNUAL/RECENT DEVELOPMENT RMSE (SGD); THE SUPPLIED-MACRO ROW USES A DIFFERENT INFORMATION SCENARIO.

Archived pipeline	Annual	Recent
History	451.1810	442.9025
Supplied-macro	451.2081	442.8892
Tree/Transformer + PLR	451.0295	441.8345

B. Evidence locations

Paths below are relative to the repository root. Model scores in the paper are local project results; method references describe algorithms rather than supplying those scores.

- Data audit: `reports/train_test_eda/summary.json` and `reports/auxiliary_data_eda/summary.json`.
- Common inputs: `experiment/03_model_comparison/same_input_models.csv` and `recommended_models.csv` in the same directory.
- Feature ladder: `experiment/02_feature_engineering/controlled_feature_ladder.csv`; group deletions: `group_feature_effects.csv` in the same directory.
- Submitted pipeline and Kaggle provenance: `experiment/01_final/submission_selection.json`. The recorded screenshot source was not available for fresh visual verification.
- Grouped blending and PLR comparisons: `experiment/04_optimization_history/09_grouped_blending.md` and `10_plr_transformer_final.md` in the same directory.
- Matched TabPFN cohort study: `reports/tabpfn_anniversary_history_80_20/`; tree/TabPFN comparison: `reports/zch_anniversary_tabpfn_blend/`.
- Cleaning and category grouping: `reports/half_median_zch_tabpfn_ensemble/` and `reports/flat_model_grouping_ensemble/`.

C. Practical reproduction sequence

Create the core environment from `environment.yml`. Use `docs/guides/running_experiments.md` for the dependency order; later rounds reuse earlier trained predictors and out-of-time predictions. The preflight command

```
python -m analysis.main preflight
-profile final
```

checks required final-model resources without training. Missing ignored model artifacts must be restored or rebuilt; running only the final blending script is insufficient.
The common-input comparison is exposed through `analysis.main common-models`. The controlled ladder uses `analysis.experiments.report_selection_trials ladder`, separately for `-stage outer` and `-stage june`. On main, `run_tabpfn_best.py` prepares the isolated TabPFN runtime; `run_tabpfn_ensemble.py` refits validation/full-data category variants and exports checked submissions. Exact member scores and blend weights are in `reports/final_report/ensemble_member_scores.csv`; submission identities are in `reports/submissions/tabpfn_ensemble_20261106/publication.json`.
The paper's tables, plots and evidence hashes are regenerated without model fitting by

```
python reports/final_report/build_report.py
```

Compile `reports/final_report/final_report.tex` with IEEEtran. The audit distinguishes recomputed metrics from archival scores; new test submissions remain unscored.

APPENDIX B
SUPPLEMENTARY VALIDATION DIAGNOSTICS

These supplementary results retain the main text's information boundaries. No additional model is fitted, and all recent-window comparisons use the same 28,995 original validation rows.

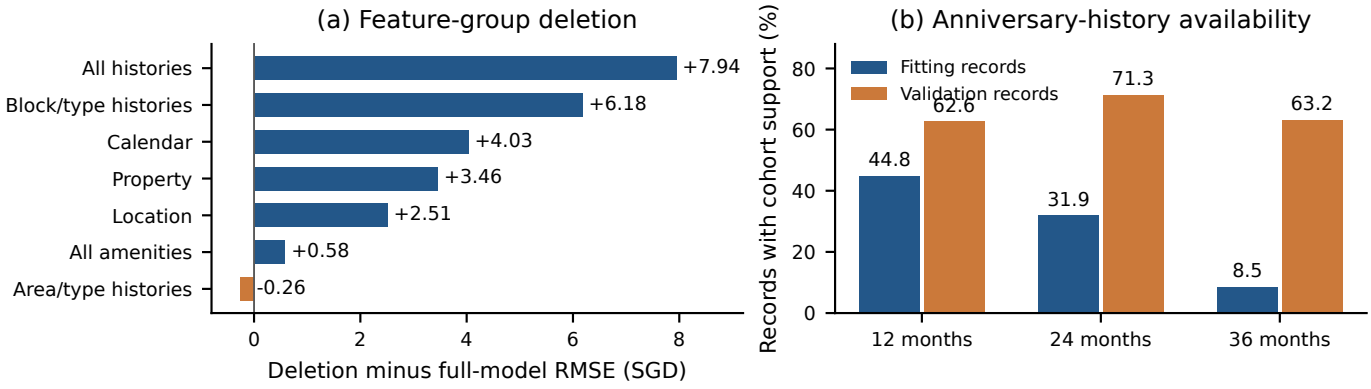


Fig. 4. Feature evidence. Left: fixed 27-input LightGBM group deletions, pooled over the three annual windows from April 2022–March 2025. Positive differences mean removal worsens RMSE; groups overlap and effects are not additive. The joint time-plus-history deletion, omitted from this scale, increases error by S$141.33. Right: cohort support in the separate TabPFN study (121,005 fitting and 28,995 validation records). Validation histories remain frozen at May 2024. Sparse fitting support and later query support need not match.

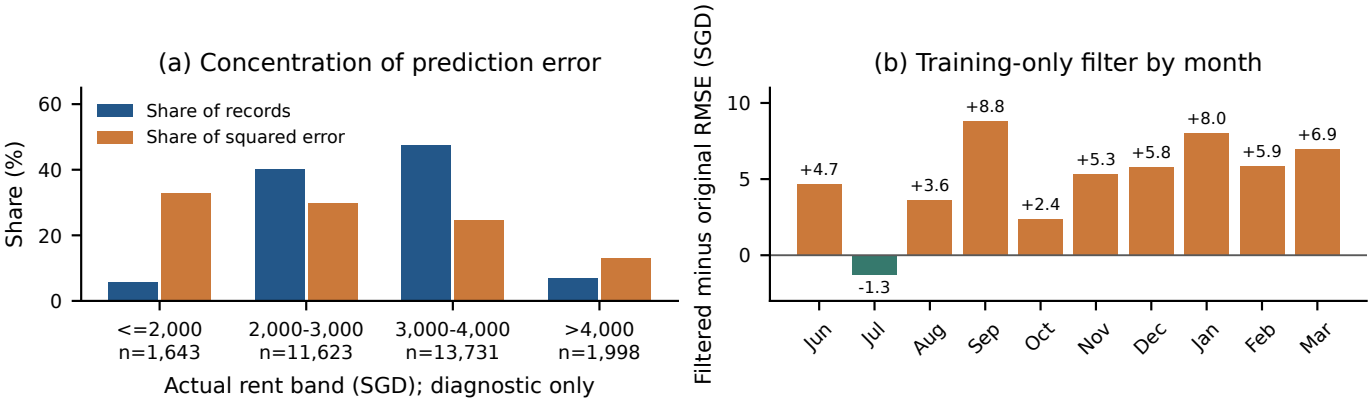


Fig. 5. Error concentration and cleaning sensitivity for the equal seven-tree/87-input TabPFN blend. Left: actual-rent groups are diagnostic only, with upper-inclusive boundaries and sample counts shown. The low-rent tail contributes far more squared error than its sample share. Right: the training-only half-median filter worsens RMSE in nine of ten June 2024–March 2025 months. All validation rows remain included. A single origin confounds horizon with calendar month.

TABLE IX
PAIRED RECENT-WINDOW RMSE SENSITIVITY (SGD); NEGATIVE VALUES FAVOR THE FIRST CANDIDATE. INPUT COUNTS FOLLOW TABLE IV. INTERVALS USE 5,000 CIRCULAR THREE-MONTH-BLOCK RESAMPLES OF TEN MONTHS, SEED 42. SAMPLED MONTHS RETAIN ALL THEIR ROWS; POOLED SQUARED ERRORS AND COUNTS DETERMINE EACH RMSE.

Contrast	ΔRMSE	95% block interval	Months improved
87 vs 77 inputs	-13.437	[-16.273, -10.991]	10/10
92 vs 87 inputs	+0.057	[-0.542, +0.773]	4/10
Equal blend vs 87	-1.007	[-1.527, -0.567]	10/10
Earlier-locked blend vs 87	+0.343	[-0.656, +1.156]	2/10
Coarser categories vs original	-0.181	[-0.499, +0.143]	6/10
Half-median filter vs original	+4.891	[+3.170, +6.461]	1/10

The cohort gain is consistent across months; coarser-category and 36-month additions remain uncertain. These descriptive intervals condition on fixed predictions and omit model-fitting and adaptive-selection uncertainty. Ten reused months provide limited temporal support and do not constitute an independent test.

APPENDIX C
TABPFN FINE-TUNING EXPERIMENTS

These experiments adapt TabPFN-3.5-Fast with the original 77 categorical-month inputs and residual target, before the anniversary-feature ablation in Table IV. Every outer score uses the same 28,995 June 2024–March 2025 transactions. Full-context inference uses all 121,005 rows through May 2024. Nine categorical positions are explicit during adaptation and inference; earlier runs with inconsistent handling are excluded.

A. Adapter and decoder comparisons

Adapters fit on 102,829 rows through November 2023; 600 sampled, population-weighted December 2023–May 2024 queries select checkpoints. The common recipe uses seed 42, AdamW with weight decay 0.01, BF16, 1,024-row tasks, two training/checkpoint views, an eight-epoch cap and patience three. DiagScale trains diagonal attention scaling; rank-8 LoRA updates Q/K across all attention modules and rank-2 LoRA only within eight in-context-learning (ICL) blocks, both at LR 0.0003. Native decoder tuning (LR 0.00003) updates the pretrained distribution head, distinct from an exported-embedding MLP. Other pretrained weights remain frozen.

TABLE X
TABPFN ADAPTATION ON THE RECENT DEVELOPMENT WINDOW (SGD). DIFFERENCES USE THE FROZEN CONTROL WITH THE SAME INFERENCE-VIEW COUNT. LR DENOTES LEARNING RATE. THE TWO MATCHED DIAGSCALE LOSS TRIALS SHARE LR 0.01 AND A REVISED, TYPED ONE-VIEW CHECKPOINT RULE.

Model / setting	Views	Updated weights	RMSE	MAE	ΔRMSE
Frozen control	1	0	455.814	329.748	+0.000
DiagScale, LR 0.003	1	8,576	456.133	331.187	+0.320
DiagScale, LR 0.005	1	8,576	457.057	331.798	+1.244
DiagScale, LR 0.01	1	8,576	455.212	330.625	-0.602
DiagScale, LR 0.05	1	8,576	460.074	338.544	+4.261
DiagScale, matched CRPS+MSE	1	8,576	455.122	330.872	-0.692
DiagScale, matched MSE only	1	8,576	455.071	330.790	-0.743
ICL Q/K LoRA, rank 2	1	65,536	455.472	330.472	-0.341
Native regression decoder	1	9,320,328	455.990	331.042	+0.176
Frozen control	2	0	455.396	327.794	+0.000
All-attention Q/K LoRA, rank 8	2	299,008	457.043	333.378	+1.647

Corrected LR 0.01 DiagScale lowers RMSE by S$0.602 but raises MAE. Lower rates do not help; LR 0.05 worsens RMSE despite the best earlier calibration score. Matched MSE-only tuning reaches 455.071, only S$0.051 below its CRPS+MSE control; their saved paired month-bootstrap interval is [-0.380, +0.298]. Comparison with the older 455.212 adapter additionally changes checkpoint selection and initialization seeding. Native decoder tuning and rank-8 LoRA lose to their frozen controls. These runs offer limited evidence of an adaptation benefit across forecast origins.

B. Chronological fine-tuning support

A separate experiment replaces mixed-month tasks with month-coherent queries and strictly earlier support. Strategies share identical query IDs, 820 support and 204 query rows per episode, 100 episodes per epoch, LR 0.01 and eight completed epochs. Uniform/recency use patience eight to match the update budget; local/mixed use patience three. The mixture is 40% local, 35% recent and 25% diverse. Query IDs and duplicate transaction groups are excluded from support.

TABLE XI
MONTH-COHERENT DIAGSCALE TRAINING: RMSE (SGD) UNDER TWO INFERENCE CONTEXTS. THESE FOUR ADAPTERS HAVE EQUAL UPDATE BUDGETS. THE ORIGINAL ADAPTER USES DIFFERENT QUERY EPISODES AND IS A DEPLOYMENT REFERENCE RATHER THAN A SUPPORT-ONLY CONTROL.

Training support	Full 121,005-row inference	Mixed 10,000-row inference
Uniform	459.627	460.786
Recency	459.063	460.819
Local/property	457.240	460.929
Mixture	457.709	459.324

Local support is best among the new full-context adapters (457.240), yet loses by S$2.028 to the original 455.212 adapter. At mixed 10,000-row inference, the new mixture adapter scores 459.324 versus 456.390 for the original adapter. Changing inference context alone is separate from fine-tuning: the latter original-adapter result beats uniform 10,000-row inference (461.266), yet loses to full context. No new support strategy improves the deployment reference.

These comparisons reuse a development period with one training seed and tasks much smaller than deployment context. They cannot rule out better adaptation recipes. Evidence is under `reports/tabpfn_components_80_20/`, `reports/tabpfn_diagscale_learning_rates_80_20/` and `reports/tabpfn_context_sampling/`; the builder rechecks labels, dates, IDs and RMSE without retraining.

APPENDIX D
SMALL TRANSFORMERS TRAINED FROM SCRATCH VERSUS TABPFN

This study asks whether a compact, task-trained Transformer can match a pretrained tabular model on the same recent rental forecasting window. It is separate from the PLR-FT channel in the main grouped pipeline. All scratch models begin with random weights, use the original labels, and predict individual-rental residuals relative to the inherited temporal baseline. Table XII reports three independent training seeds; its mean is the arithmetic mean of per-seed RMSE, not the error of an ensemble of those seeds.

A. Architecture and chronological training

T0 encodes the 77 property/history inputs with two feature-attention blocks, width 128, four heads, dropout 0.1 and spatial Fourier features. It has 1,729,185 trainable parameters. T1 adds a 12-month rental-sequence branch with one attention block, width 32 and two heads; T2 adds COE channels and T3 adds the supplied median Singapore-stock-return proxy. These sequence models have 1,739,331 parameters each. They are small supervised models trained on this dataset, with no foundation-model pretraining.

All rent-derived inputs use the row's forecast cutoff; macro inputs have an additional one-month buffer. Training episodes cover horizons 1–16. The inherited rent baseline plateaus after month ten, while the horizon input continues increasing. AdamW uses learning rate 0.0003, weight decay 0.001, batch size 256, standardized residual MSE and gradient clipping at one. A 30-epoch cap and patience five apply to earlier calibration; the rounded median best epoch across September 2022–June 2023 and July 2023–April 2024 selects each variant's outer training length. Each earlier window contains 2,000 population-weighted query cases. Later validation labels do not select those epochs.

TABLE XII
SCRATCH TRANSFORMER RMSE (SGD) ON THE SAME 28,995 JUNE 2024–MARCH 2025 ROWS. THE LAST ROW BLENDS INDEPENDENTLY TRAINED T0/T1 PREDICTIONS USING A DECAY CHOSEN ON EARLIER DATA.

Model	Seed 42	Seed 43	Seed 44	Mean RMSE
T0: property/history Transformer	462.336	459.415	461.740	461.164
T1: T0 + rent sequence	461.901	465.875	468.151	465.309
T2: T1 + COE sequence	470.856	466.344	463.130	466.777
T3: T2 + stock-return proxy	465.960	467.068	462.230	465.086
Earlier-selected decayed T0/T1 blend	461.893	458.612	460.119	460.208

TABLE XIII
TABPFN REFERENCES ON THE IDENTICAL RECENT-WINDOW ROWS (SGD). CDE IS THE EARLIER-SELECTED FAMILY; ABCDE EQUAL IS AN EQUAL-WEIGHT SENSITIVITY COMBINING AN ADAPTER AND FROZEN FEATURE VIEWS. BOTH DIFFER FROM THE LATER ANNIVERSARY MODEL. THESE ARE COMPLETE-PIPELINE COMPARISONS, NOT CONTROLLED TESTS OF PRETRAINING ALONE.

Reference pipeline	Views	RMSE	MAE
Frozen TabPFN, original 77 inputs	2	455.396	327.794
TabPFN CDE, earlier-selected family	2	455.360	329.033
TabPFN ABCDE equal ensemble	varied	454.704	328.526

B. Results and interpretation

T0 averages 461.164 RMSE, S$6.460 above the TabPFN ABCDE equal ensemble (454.704) and S$5.768 above frozen two-view TabPFN (455.396). The additional sequence branches raise mean RMSE to 465.086–466.777. Extra temporal channels therefore do not improve this scratch-model design, despite causal feature construction. Only 51 labelled calendar months are available; the large transaction count does not supply an equally large number of independent market regimes. Limited temporal support and overlapping engineered histories are plausible explanations, rather than isolated causes.

The earlier-selected forecast blend is

$$\widehat{y}_{\mathrm{decay}}(h) = \widehat{y}_{T0}(h) + 0.5 \exp\left(-\frac{h-1}{3}\right) \left[\widehat{y}_{T1}(h) - \widehat{y}_{T0}(h)\right].$$

It lowers mean RMSE to 460.208 and improves all three seeds, but remains S$5.504 above ABCDE. On a separate 16-month diagnostic its RMSE changes from 492.408 to 492.554, so the gain does not extend uniformly to the longer window. Because T0 and T1 are trained independently, their forecast difference is not an isolated temporal-branch contribution. A direct internal decay gate failed earlier transfer and did not advance to outer evaluation.

The newer frozen TabPFN with ten anniversary fields reaches 441.959 (Table IV), S$19.205 below mean T0. That larger gap also changes the input representation; it cannot be attributed entirely to model size or pretraining. The present evidence favors the tested TabPFN pipelines over these small scratch Transformers, while repeated development-window reuse prevents a confirmatory generalization claim. The scratch series was not promoted to a submission.

Architecture, split and training settings are in `reports/spatial_temporal_transformer/settings.json`; per-seed predictions and metrics are in the same directory. The decay comparison is under `reports/transformer_ranked_experiments/23_forced_decay/`, and the TabPFN reference is under `reports/tabpfn_abcde_ensemble/`. Generated `reports/final_report/supplementary_neural_scores.csv` and the evidence audit record independently recomputed results for both new appendices.

APPENDIX E
INTERPRETING TABPFN'S PREDICTIVE ADVANTAGE

Appendix D shows an advantage for the tested TabPFN pipelines over the small scratch Transformers. The mechanisms below are plausible explanations, supported to different degrees by the literature and our ablations. Our experiments compare complete systems and do not identify the causal contribution of pretraining alone.

A. A learned prior reduces what must be learned locally

TabPFN is pretrained across synthetic tabular prediction tasks, learning an inference procedure before seeing the target dataset [11]. Its prior encodes preferences over possible feature–target relationships; the prior-data-fitting objective approximates Bayesian prediction under the synthetic task distribution. This can regularize prediction when local evidence is sparse. We hypothesize that this helps with unevenly supported property cohorts. T0 instead learns its representations and prediction rule from this rental dataset alone. Although it has 121,005 fitting transactions, they span only 41 calendar months, limiting independent evidence about market changes. The prior's suitability for this task is an interpretation, rather than a measured pretraining effect.

B. In-context learning uses labelled examples at prediction time

Frozen weights still permit dataset-specific adaptation: TabPFN conditions a query prediction on the feature–label pairs in its supplied context, without gradient updates [11]. Our scratch model attends across a property's feature tokens and optional monthly sequence; it does not receive the labelled transaction table as inference context. Both methods use fitting labels, but through different computation paths. TabPFN's context can therefore support query-dependent use of historical examples. This is learned inference, and attention weights alone would not establish that it retrieves economically meaningful comparable flats.

TABLE XIV
LOCAL EVIDENCE RELEVANT TO THE PROPOSED MECHANISMS. ALL ERRORS ARE RECENT-WINDOW RMSE IN SGD. THESE COMPARISONS SUPPORT INTERPRETATION WITHOUT ISOLATING A SINGLE CAUSE.

Proposed contribution	Observed evidence	What remains unresolved
Useful labelled context	Fixed DiagScale: full context 455.212; uniform 10k context 461.266; mixed 10k context 456.390.	Context size and composition matter in this setup; this does not isolate pretraining.
Effective frozen inference	Frozen two-view TabPFN scores 455.396 versus mean T0 461.164; rank-8 LoRA worsens TabPFN to 457.043.	Gradient adaptation is unnecessary for this observed advantage; its failure does not prove the prior is optimal.
Explicit rental structure	Keeping TabPFN frozen, adding 12/24-month cohort fields lowers RMSE from 455.396 to 441.959.	This directly supports the representation change; it is not an architecture-only gain.

C. Tabular representations, capacity and task formulation

The 3.5 report describes distribution-aware cell encodings, including column ranks and learned Fourier features, and a synthetic prior emphasizing high-cardinality and grouped data [10]. These designs may help with heterogeneous numeric and categorical inputs. It also lists roughly 84 million parameters for Fast, compared with T0's 1.73 million. Capacity, encodings and prior training are therefore confounded here. We have no ablation showing which encoding or parameter count explains the rental result. Frozen TabPFN still relies on our chronological histories and temporal residual baseline; its success does not demonstrate autonomous discovery of rental dynamics or reliable extrapolation beyond the validated horizon.

D. What would distinguish these explanations?

A stronger test would compare pretrained and randomly initialized versions of the same backbone, using matched inputs and a declared training budget, across new chronological origins. Fixed-checkpoint context-size and sampling ablations would separately test reliance on support examples. These are proposed follow-ups, not completed experiments. For now, the evidence supports a useful combination of pretrained inference, labelled context and explicit rental features; it does not establish universal TabPFN superiority or immunity to temporal shift.

APPENDIX F
AUXILIARY TEMPORAL-DATA EXPERIMENTS

This appendix evaluates the supplied Certificate of Entitlement (COE) auction and stock-price datasets as temporal predictors. COE may proxy changing economic conditions, while stock returns may reflect business activity; these are predictive hypotheses rather than causal channels established by our data. The studies below distinguish individual-rental error from national or flat-type mean-rent error, and distinguish refreshed forecasts from forecasts made once at a fixed origin.

A. Direct economic features in a controlled tree model

The first study holds the 29-input property/time/block/amenity baseline and HistGradientBoosting configuration fixed (200 iterations, seed 42). Six COE fields add category A/C monthly median premiums, growth and bids-to-quota ratios. Three stock fields add the median Singapore-listed adjusted-close return, compounded three-month return and rolling volatility. Thus the combined variant has 38 inputs. Approval-month inputs end in the preceding month: this is a monthly-updated information scenario, conditional on publication, rather than one forecast from a common cutoff.

TABLE XV
DIRECT AUXILIARY-FEATURE ABLATION: INDIVIDUAL-RENTAL RMSE (SGD). EXPANDING-WINDOW VALIDATION ENDS IN DECEMBER 2022, JUNE 2023, MARCH 2024 AND MARCH 2025, RESPECTIVELY. FEATURE COUNTS ARE IN PARENTHESES. EACH ROW COMPARES IDENTICAL FITTING/VALIDATION RECORDS AND MODEL SETTINGS.

Validation start	Rows	29-input control	+ COE (35)	+ Stocks (32)	+ Both (38)
2022-07	15175	658.14	658.64	660.93	661.98
2023-01	17754	642.74	642.94	645.32	645.39
2023-07	26905	567.69	567.45	567.51	567.99
2024-04	35158	478.82	479.63	479.54	479.92

COE and stock bundles each improve only one of four windows; their combination worsens all four. A separate decomposition of COE premiums alone found gains below S$0.50 in earlier windows and a S$0.14 loss in the latest window. These small, inconsistent differences do not support a default addition of realized economic inputs.

Sector-specific follow-ups also fail to transfer. Earlier windows select engineering-stock features for the individual-rental tree; they raise recent RMSE from 471.631 to 472.168 with monthly updates and to 472.196 with fixed-origin stock values. In a separate flat-type mean-rent prototype, adding COE to pooled Ridge raises count-weighted RMSE from 42.987 to 80.842 at the June 2024 origin, and from 227.206 to 402.729 at the November 2023 origin. Those overlapping holdouts measure group means, not individual rentals.

B. Learning national rental dynamics from sequences

A second study compares persistence, momentum, EWMA, lag/summary Ridge, MLP and LSTM over rent, rent+COE, rent+stocks and joint histories. The 43 specifications per target use only observations and mature forecast labels available at each origin. Neural models use eight hidden units, 100 fixed epochs, seed 42 and direct multi-horizon outputs. History windows of 3/6/12 months are chosen on earlier data. The target below is the property-adjusted national arithmetic mean, using a frozen 2021 property reference; it remains sensitive to changes in property premiums.

TABLE XVI
PROPERTY-ADJUSTED NATIONAL-MEAN RMSE (SGD), JUNE 2024–MARCH 2025. FIXED-ORIGIN FORECASTS ALL TEN MONTHS USING MAY 2024 INFORMATION. ROLLING POOLS 1/3/6-MONTH FORECASTS, AVERAGING SQUARED ERRORS WITHIN EACH TARGET MONTH BEFORE AVERAGING MONTHS. EACH LSTM INPUT SET USES ITS OWN EARLIER-SELECTED WINDOW.

Input / processor	Fixed-origin RMSE	Rolling RMSE
Rent persistence	45.388	37.746
LSTM: rent only	258.483	126.409
LSTM: rent + COE	23.244	44.958
LSTM: rent + stocks	133.216	125.095
LSTM: rent + stocks + COE	31.828	61.503

Rent+COE substantially improves this LSTM's fixed path, and adding stocks does not improve on that information set. However, persistence wins the earlier overall model selection and beats every displayed LSTM in rolling evaluation. Rent+COE uses twelve history months for the fixed path and six for rolling; the joint LSTM uses three. These are input-plus-window comparisons, not fixed-window estimates of COE's incremental effect. An extra month of input delay worsens fixed-origin rent+COE RMSE from 23.244 to 49.118; the delayed persistence control scores 54.521. Availability assumptions materially affect the result.

C. Transfer from national forecasts to individual rentals

An additive correction transfers predicted movement in the adjusted national index to the frozen 29-input tree. Its scale is constrained to zero–one and fitted only on July 2023–May 2024 rental errors. The same 28,995 June 2024–March 2025 rentals evaluate every correction. Rolling corrections can use newly observed rents after the tree fit; fixed-origin corrections use May 2024 information throughout.

TABLE XVII
INDIVIDUAL-RENTAL RMSE (SGD) AFTER AN EARLIER-FITTED NATIONAL-INDEX CORRECTION. BOTH SCENARIOS START FROM THE SAME UNCORRECTED TREE RMSE OF 471.631. NEGATIVE DIFFERENCES IMPROVE THAT CONTROL. FOLLOW-UP ROWS WERE ADDED AFTER INSPECTION OF THE AGGREGATE STUDY; THEIR MODELS, WINDOWS AND SCALES STILL USE EARLIER DATA.

Correction to 471.631 tree control	Fixed RMSE	Δ	Rolling RMSE	Δ
Rent persistence	471.631	+0.000	469.618	-2.013
Rent-only LSTM (follow-up)	469.398	-2.233	470.096	-1.534
Rent + stock + COE Ridge	470.375	-1.256	470.049	-1.582
Rent + stock + COE LSTM	470.905	-0.726	470.271	-1.360
Rent + COE LSTM (follow-up)	471.228	-0.403	470.284	-1.347

The fixed-origin joint LSTM improves the tree by S$0.726, yet the rent-only LSTM follow-up improves it by S$2.233. Rolling persistence improves by S$2.013, more than the joint LSTM's S$1.360. Thus a better national-mean forecast does not necessarily improve individual-rental prediction, and the tree-correction gains cannot all be credited to external economic data. The correction can partly compensate for bias in the frozen tree. These are tree experiments; they do not measure a gain for the final tree/TabPFN ensemble.

D. Forecast-derived COE inputs in frozen TabPFN

The matched TabPFN follow-up preserves the 77 original inputs, residual target, additive temporal baseline, all 121,005 context rows, two estimator views and seed 42. It first adds five missing/sparse-history/unknown-block flags, then a rent-only national LSTM growth forecast and an unavailable flag, then the difference between rent+COE and rent-only forecast growth. Both LSTMs use twelve months, eight hidden units and 100 fixed epochs. Their source observations stop one month before each row's original forecast cutoff; unsupported early forecasts remain missing. No realized future COE values enter these forecasts.

TABLE XVIII
NESTED FORECAST-INPUT ADDITIONS TO FROZEN TABPFN: INDIVIDUAL-RENTAL ERRORS IN SGD ON THE IDENTICAL RECENT-WINDOW ROWS. DIFFERENCES USE THE ORIGINAL 77-INPUT CONTROL. THE UNAVAILABLE FLAG AND RENT FORECAST ENTER TOGETHER.

Nested inputs	Fields	RMSE	MAE	ΔRMSE
Original control	77	455.396	327.794	+0.000
+ Reliability flags	82	455.962	328.283	+0.566
+ Rent-only LSTM forecast	84	456.707	328.417	+1.311
+ Incremental COE forecast	85	456.905	329.586	+1.509

Rent-only forecast inputs win the earlier feature selection, reducing pooled earlier RMSE from 601.934 to 596.443, but raise later RMSE by S$1.311. The incremental COE field adds a further S$0.198 loss relative to that rent-forecast variant. This isolates the nested COE addition within this feature pipeline; it does not show that every economic forecasting method would fail. Sparse startup support and changing forecast distributions remain limitations. Appendix D separately reports the scratch Transformer's rent/COE/stock-sequence comparison, where added channels also fail to improve mean recent RMSE.

E. Interpretation and evidence trail

The auxiliary sources provide only 51 labelled calendar months in total, with serial dependence and changing market regimes. Many transactions share the same macro observation. Supplied adjusted-stock prices and constituents are snapshots rather than a reconstructed historical release vintage. The national stock-sequence proxy uses 22 matched historical ETF holdings, not an official STI series; it differs from the median-return basket in the direct-feature study. Monthly updating, fixed-origin prediction and the main text's supplied prediction-month macro experiment are distinct information settings.

The strongest supported finding is mixed transfer: COE helps some aggregate fixed-path forecasts, while direct features and forecast-derived TabPFN inputs do not improve consistently. Reused development windows, many tried specifications, overlapping horizons and uncertain release timing limit generalization claims. No hidden-test score or new submission is inferred from these ablations.

Evidence is retained under `reports/economic_feature_ablation/`, `reports/stock_sector_signals/`, `reports/flat_type_coe_forecast/`, `reports/temporal_sequence_ablation/` and `reports/tabpfn_reliability_macro_80_20/`. The report builder regenerates these tables and `reports/final_report/supplementary_auxiliary_scores.csv`; its audit distinguishes reconstruction from monthly summaries, saved national forecasts and label-checked TabPFN row predictions. No model is refitted during report preparation.